

Dense Expands, Sparse Anchors: Channel-Asymmetric LLM Query Expansion for Hybrid Retrieval

Chunran Zhang

School of Computing and Artificial Intelligence
Southwest Jiaotong University
Chengdu, China
chronis@my.swjtu.edu.cn

1 Introduction

LLM-based query expansion has become a common approach to improving retrieval effectiveness (Gao et al., 2023; Wang et al., 2023; Zhang et al., 2024). For hybrid retrieval, its gains are typically measured at a preset top- L cutoff using metrics such as nDCG and recall (Zhang et al., 2024; Liu and Zhang, 2025). Recent work on fixed top- L fusion shows that L does more than control access depth: changing the cutoff can also change which candidates and cross-channel contributions enter the fused ranking (Zhang, 2026). Consequently, a gain observed at one L reflects both the query expansion and the chosen cutoff; at another L , its magnitude—and even the ordering between methods—may change. Evidence from a single cutoff therefore does not isolate the effect of query expansion.

To remove this confound, we evaluate the retrieval effectiveness of each query construction using complete-list fusion, without truncating either channel (Zhang, 2026). We then reveal the dense and sparse rankings progressively and record the per-channel depths at which the same ordered top- K becomes fixed. This separates two questions: what fused ranking does an expansion produce, and how much ranked evidence is required to determine it? The answers need not move together. An expansion may improve retrieval effectiveness while pushing decisive evidence deeper into one channel. We therefore ask whether query expansion can improve effectiveness without increasing the access depth of either channel.

Meeting both goals requires treating dense and sparse retrieval differently. Prior studies show that expansion gains vary across retrievers, and that dense and sparse retrieval can favor different forms of generated evidence (Weller et al., 2024; Li et al., 2026). Our NFCorpus analysis shows what happens when one shared expansion is applied to both

channels. When the same generated passages are used in both channels, retrieval effectiveness improves and fewer dense ranks are needed to determine the fused result, yet more sparse ranks are required. The unexpanded sparse query matches only a small portion of the corpus; the added terms make far more documents eligible for sparse ranking, so the list no longer exhausts early. The same generated content therefore supplies useful semantic evidence to the dense retriever while spreading lexical evidence across a much larger sparse list. A shared expansion cannot control these two effects separately.

We address this asymmetry with channel-asymmetric query expansion. The LLM generates several complementary reference passages around the original query. For dense retrieval, orthogonal residual expansion preserves the original query direction and adds only the new semantic component supplied by the references. For sparse retrieval, score-product anchoring multiplies each document’s scores under the unexpanded and expanded queries. Documents matched only by generated terms receive no anchored score, while the new terms can still reorder documents supported by the original query. Together, these operations broaden semantic coverage without broadening sparse support: Dense expands; sparse anchors.

The evaluation mirrors this argument. A top- L sensitivity study first examines whether the measured gains and relative ranking of query expansion methods change with the cutoff. We then evaluate complete-list fusion on seven public retrieval benchmarks, comparing our method with the unexpanded query, shared expansion, and representative generative expansion methods in terms of retrieval effectiveness and per-channel access depth. Controlled experiments examine the contributions of complementary reference generation, orthogonal residual expansion, and score-product anchoring, followed by robustness analyses across generation

and retrieval settings. Together, these experiments test whether channel-asymmetric query expansion can produce better fused rankings while reducing the access required from both channels.

References

- Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. 2023. [Precise zero-shot dense retrieval without relevance labels](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1762–1777, Toronto, Canada. Association for Computational Linguistics.
- Chenyi Li, Xinhui Tu, and Zaixiang Wang. 2026. [DVCQR: Dual-view conversational query rewriting with stage-wise reinforcement learning](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 22993–23014, San Diego, California, United States. Association for Computational Linguistics.
- Lingyuan Liu and Mengxiang Zhang. 2025. [Exp4Fuse: A rank fusion framework for enhanced sparse retrieval using large language model-based query expansion](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 163–173, Vienna, Austria. Association for Computational Linguistics.
- Liang Wang, Nan Yang, and Furu Wei. 2023. [Query2doc: Query expansion with large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9414–9423, Singapore. Association for Computational Linguistics.
- Orion Weller, Kyle Lo, David Wadden, Dawn Lawrie, Benjamin Van Durme, Arman Cohan, and Luca Soldaini. 2024. [When do generative query and document expansions fail? a comprehensive study across methods, retrievers, and datasets](#). In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 1987–2003, St. Julian’s, Malta. Association for Computational Linguistics.
- Chunran Zhang. 2026. [Exact adaptive hybrid retrieval without fixed top-l cutoffs](#). *Preprint*, arXiv:2608.07152.
- Le Zhang, Yihong Wu, Qian Yang, and Jian-Yun Nie. 2024. [Exploring the best practices of query expansion with large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1872–1883, Miami, Florida, USA. Association for Computational Linguistics.